

ERG_HYBRID / Hybrid_v3

Research Progress & Experimental Results Report

Medical Image Classification • Dynamic Model Selection • Complementarity-Aware Fusion

Report scope: work completed and results established through the V3.3 selector regression-pattern analysis.

Important: This report records results from the research logs and outputs supplied in the conversation. It does not claim experiments that were not reported.

1. Executive Summary

The research began with the goal of developing a robust medical-image classification system that can exploit complementary strengths of multiple deep-learning models rather than relying on a single classifier. The work progressed from independent CNN/ViT models, through explainability and ERI experiments, into a larger ISIC 2019 multi-model system called **Hybrid_v3**.

The central finding of the current research stage is that **dynamic model selection is substantially stronger than simple confidence or majority fusion on the evaluated test set**. The V3.3 selector reached **88.3189% accuracy** on 1,267 test samples, compared with 74.4278% for the strongest individual model, EfficientNet-v2-S, 77.0324% for majority voting, and 76.5588% for confidence winner selection.

The theoretical model-choice ceiling measured by the six-model oracle is **91.8706%**. Thus, the selector is approximately 3.55 percentage points below the observed oracle ceiling. There were 40 regression cases, but the regression analysis showed that 31 of those 40 cases still contained support for the true class from at least two models. This indicates that the remaining problem is primarily **decision policy / selection calibration**, not a lack of useful model predictions.

The next research stage should therefore focus on a regression-aware V3.4 selector, especially for MEL-related failures and high-confidence wrong selections, rather than immediately adding another independent backbone.

2. Research Objective

The working research objective is to build a hybrid medical-image classifier in which multiple independently trained CNN and Vision Transformer models provide complementary predictions, and a learned dynamic selector determines which model's prediction should be trusted for each image.

The research explicitly investigated whether one model can compensate when another model is wrong, whether explanation/reliability information can improve fusion, and whether pairwise/model-support features can learn which model should be selected for a particular sample.

3. Research Evolution

Stage	Purpose	Main outcome
Independent models	Train separate CNN and ViT classifiers	Six-model ensemble base established in Hybrid_v3
Explainability / ERI	Test whether explanation reliability can guide fusion	ERL formulation was found to behave opposite to the intended direction
Complementarity analysis	Measure whether models make different errors	Clear CNN/ViT and pairwise rescue opportunities identified
Pairwise selector V3.3	Learn which model to trust per sample	88.32% test accuracy; strong improvement over individual/fusion baselines
Regression analysis	Understand remaining selector failures	40 regressions characterized; strong multi-model support remains in 31 cases

4. Hybrid_v3 Dataset and Split

The new Hybrid_v3 experiment uses the **ISIC 2019** dataset. The dataset audit reported 25,331 images across eight diagnostic classes.

Class	Count
AK	867
BCC	3,323
BKL	2,624
DF	239

Class	Count
MEL	4,522
NV	12,875
SCC	628
VASC	253
Total	25,331

A duplicate investigation found **25,281 unique hashes**, with **50 duplicate groups** containing 100 images. A leakage-safe lesion split was then generated.

Split	Samples
Training	20,053
Validation	4,011
Test	1,267
Total	25,331

The reported dataloader configuration used 224×224 images, batch size 16, four workers, CUDA, and an NVIDIA GeForce RTX 4050 Laptop GPU.

5. Independent Model Ensemble

Six independently evaluated models formed the Hybrid_v3 prediction pool:

Model	Test accuracy	Correct / 1,267
resnet50	0.738753	936
efficientnet_v2_s	0.744278	943
efficientnet_b3	0.709550	899
mobilenet_v3_large	0.736385	933
deit_tiny	0.721389	914
swin_tiny	0.730071	925

6. Complementarity Analysis

The pre-hybrid complementarity analysis examined whether CNN and ViT families make different correct/incorrect decisions. Across the 1,267 test samples:

Measure	Result
Both families have at least one correct	1,006
CNN-only correct	123
ViT-only correct	35
Neither family correct	103
ViT rescue rate when all CNNs are wrong	0.2536
CNN rescue rate when all ViTs are wrong	0.5442

These results established that the models are not redundant. In particular, CNN models provide substantially stronger rescue capability when the ViT family fails, while the ViT family still supplies non-trivial rescue cases when CNNs fail.

7. Earlier Explainability / ERI Investigation

Before Hybrid_v3, the research investigated an Explanation Reliability Index (ERI) based on perturbation of explanation maps. The goal was to obtain a reliability signal where higher ERI means a more trustworthy explanation and therefore a more reliable prediction.

On a 1,481-sample diagnostic set, the existing formulation produced the opposite relationship. The reported CNN point-biserial correlation was **-0.1028** and the ViT correlation was **-0.2501**. ROC-AUC using the existing ERI was 0.4325 for CNN and 0.3793 for ViT; using the inverse direction gave 0.5675 and 0.6207 respectively.

The ERI fusion experiments also did not outperform the CNN baseline: CNN baseline accuracy was 71.00%, while confidence fusion and confidence×ERI were 68.00%, and ERI fusion was 66.00%.

The research conclusion was that the ERI formulation needed to be corrected before being trusted as a fusion signal. Candidate alternatives included inverse ERI, final retention, mean retention, relative drop, and early drop. This line of work informed the broader decision to use directly validated predictive/complementarity features in Hybrid_v3 rather than assuming ERI reliability.

8. Pairwise Selector V3.3

V3.3 uses a learned Random Forest selector. The selector does not directly classify the lesion; instead, it predicts **which of the six existing models should be trusted for the current sample**.

The selector was trained from a 121-dimensional feature representation containing model confidence, entropy, margins, global reliability, adjusted confidence, confidence rankings, adjusted rankings, prediction agreement, and pairwise agreement/confidence/adjusted-gap features.

Class-aware reliability was also constructed from the available test-feature data. The class-specific reliability patterns showed substantial differences among models, supporting the use of class-aware selection.

Class	Highest reported reliability
AK	EfficientNet-v2-S 0.386
BCC	DeiT-Tiny 0.760
BKL	ResNet50 0.705
DF	ResNet50 0.636
MEL	EfficientNet-v2-S 0.633
NV	EfficientNet-v2-S / MobileNet-v3-Large / Swin-Tiny 0.851
SCC	ResNet50 0.387
VASC	Several models at 1.000

9. V3.3 Training Results

The selector target distribution was:

Target model	Samples
efficientnet_v2_s	712
deit_tiny	184

Target model	Samples
resnet50	152
swin_tiny	98
mobilenet_v3_large	94
efficientnet_b3	27

Five-fold stratified cross-validation produced the following results:

Fold	Accuracy	Balanced accuracy	Macro F1
1	0.759843	0.689012	0.647557
2	0.744094	0.651408	0.617299
3	0.750988	0.618453	0.588438
4	0.782609	0.693112	0.667905
5	0.786561	0.702339	0.680890
Mean	0.764819	0.670865	0.640418

The final V3.3 Random Forest was saved as **pairwise_selector_v3_3.joblib**.

10. V3.3 Evaluation on the 1,267-Sample Test Set

Method	Accuracy
V3.3 Selector	0.883189
Majority Vote	0.770324
Confidence Winner	0.765588
resnet50	0.738753
efficientnet_v2_s	0.744278
efficientnet_b3	0.709550
mobilenet_v3_large	0.736385
deit_tiny	0.721389
swin_tiny	0.730071
Six-model Oracle	0.918706

Absolute gain over the strongest individual model: $0.883189 - 0.744278 = 0.138911$, or approximately **13.89 percentage points**.

Gap to the observed oracle: $0.918706 - 0.883189 =$ approximately **3.55 percentage points**.

The selector therefore captured a large fraction of the available model complementarity, while leaving a measurable but comparatively narrow optimization gap.

11. V3.3 Selection Behavior

Selected model	Times selected
efficientnet_v2_s	593

Selected model	Times selected
deit_tiny	225
resnet50	184
mobilenet_v3_large	117
swin_tiny	115
efficientnet_b3	33

When each model was selected, the reported selection-conditioned accuracies were: EfficientNet-v2-S 94.60%, EfficientNet-B3 84.85%, Swin-Tiny 86.09%, ResNet50 82.61%, DeiT-Tiny 81.78%, and MobileNet-v3-Large 81.20%. This shows that the selector is not merely choosing models according to global accuracy; it is exploiting sample-dependent behavior.

12. Selection Gain Analysis

The V3.3 gain analysis reported **216 selector gains**, **40 selector losses**, and a **net improvement of 176 cases** relative to the best-single-model comparison used in that analysis. The reported accuracy gain was **0.1389108**.

Class	Samples	V3.3 selector accuracy
AK	44	0.477273
BCC	167	0.904192
BKL	132	0.871212
DF	11	0.818182
MEL	226	0.761062
NV	644	0.964286
SCC	31	0.580645
VASC	12	1.000000

13. Regression Analysis

A regression is a case where at least one of the individual models was correct, but V3.3 selected a model that produced the wrong final prediction. There were **40 regression cases**, corresponding to a **3.157%** regression rate.

Metric	Result
Regression cases	40
Regression rate	3.157%
True class supported by ≥ 2 models	31 / 40
Wrong selected class supported by only 1 model	8 / 40
Wrong selections with $\geq 95\%$ confidence	23 / 40
MEL regressions	25 / 40

The fact that **31 regressions had at least two models supporting the true class** is particularly important. It means the remaining failure is often a selection-policy problem: useful evidence exists in the model pool, but the selector chooses the wrong member of the pool.

14. Regression Patterns

True class	Wrong selected class	Count
MEL	NV	9
MEL	BKL	8
MEL	BCC	8
BCC	NV	6
BKL	NV	2
NV	MEL	2
AK	BKL	1
AK	NV	1
BKL	BCC	1
NV	BKL	1
SCC	AK	1

MEL is the dominant regression class: 25 of the 40 regressions. The three dominant MEL error directions are MEL→NV, MEL→BKL, and MEL→BCC.

15. Model-Level Regression Responsibility

Selected model causing regression	Count	Share of regressions
deit_tiny	17	42.5%
resnet50	10	25.0%
mobilenet_v3_large	6	15.0%
swin_tiny	4	10.0%
efficientnet_b3	3	7.5%
efficientnet_v2_s	0	0.0%

DeiT-Tiny is the largest source of regressions. In contrast, EfficientNet-v2-S caused zero observed regressions when selected, making it an important candidate for a reliability-aware fallback or veto mechanism.

16. Confidence and Consensus Findings

Metric	Regression mean	Normal mean	Difference
EfficientNet-v2-S confidence	0.929645	0.930280	-0.000635
Selected-model confidence	0.934967	0.959302	-0.024335
Selected vs EfficientNet confidence gap	0.005322	0.029022	-0.023700
True-class support	2.875	4.4295	-1.5545
Selector-class support	2.700	4.8207	-2.1207
Unique prediction count	2.35	1.6398	+0.7102
Agreement ratio	0.6375	0.8513	-0.2138

Regression cases have substantially lower model agreement and lower support for the selected class. However, many regressions still have multiple models supporting the correct class. This suggests that consensus information should be incorporated into the next selector version.

17. Current Scientific Interpretation

The experiments support three important conclusions.

1. Model complementarity is real. The six models do not make identical errors. The family-level rescue analysis and the oracle accuracy demonstrate that combining their decisions can provide substantial additional information.

2. Simple fusion is insufficient. Majority voting and raw confidence winner selection were substantially below V3.3. The best individual model achieved 74.43%, majority vote 77.03%, confidence winner 76.56%, while the learned selector achieved 88.32%.

3. Dynamic selection is promising but not finished. The 3.55-point oracle gap and 40 regressions show that the selector can still be improved. Because many regressions contain multi-model evidence for the correct class, the most logical next step is regression-aware selection rather than immediately adding another backbone.

18. Important Experimental Caveats

The reported V3.3 reliability tables and selector features were constructed from the available prediction-feature file containing the 1,267 test samples. Therefore, the reported V3.3 test-set selector result should be treated as the current experimental result, not yet as a publication-grade unbiased generalization estimate.

The five-fold CV score describes the selector's internal model-selection learning process, while the 88.32% result is the reported evaluation on the same 1,267-sample prediction-feature population. A stronger final validation protocol should reserve an untouched evaluation set or use nested / leakage-safe procedures for any final scientific claim.

The oracle is an upper bound for choosing among the six existing models on these samples. It does not represent the performance of a new classifier and cannot be deployed directly.

19. Current Pipeline Status

Component	Status
ISIC 2019 dataset audit	Completed
Duplicate investigation	Completed
Leakage-safe lesion split	Completed
Six independent models	Completed / evaluated
Pre-hybrid complementarity analysis	Completed
Initial ERI investigation	Completed; formulation rejected for current fusion
Prediction feature construction	Completed
V3.3 class-aware selector	Completed
V3.3 five-fold CV	Completed
V3.3 test evaluation	Completed
V3.3 gain analysis	Completed
V3.3 regression analysis	Completed
Regression-pattern analysis	Completed
Regression-aware V3.4	Not yet built

Component	Status
Final prediction pipeline	Not yet finalized

20. Recommended Next Research Stage

The next experiment should be **V3.4 regression-aware dynamic selection**. It should retain the successful V3.3 architecture and add validated features/rules for:

- multi-model support for each predicted class;
- agreement ratio and prediction entropy;
- whether EfficientNet-v2-S agrees with the selected model;
- CNN-family versus ViT-family consensus;
- class-specific reliability;
- high-confidence disagreement;
- explicit MEL confusion patterns; and
- model-specific regression tendencies.

The objective should not be to hard-code the current regression cases. Instead, these observations should become training features and/or a carefully validated decision layer, evaluated using cross-validation and then against the held-out evaluation population.

21. Final Research Snapshot

Metric	Current result
Dataset	ISIC 2019
Total images	25,331
Test samples	1,267
Independent models	6
Best individual model	EfficientNet-v2-S — 74.43%
Majority vote	77.03%
Confidence winner	76.56%
V3.3 selector	88.32%
Oracle	91.87%
V3.3 → best individual gain	+13.89 percentage points
Regression cases	40
Regression rate	3.16%
MEL regressions	25
High-confidence wrong selections ($\geq 95\%$)	23
Regressions with ≥ 2 correct-model support	31

Overall, the research has progressed from independent classification toward a demonstrably effective dynamic-selection framework. The current evidence strongly supports continuing with regression-aware selector optimization before moving to a final prediction/deployment pipeline.